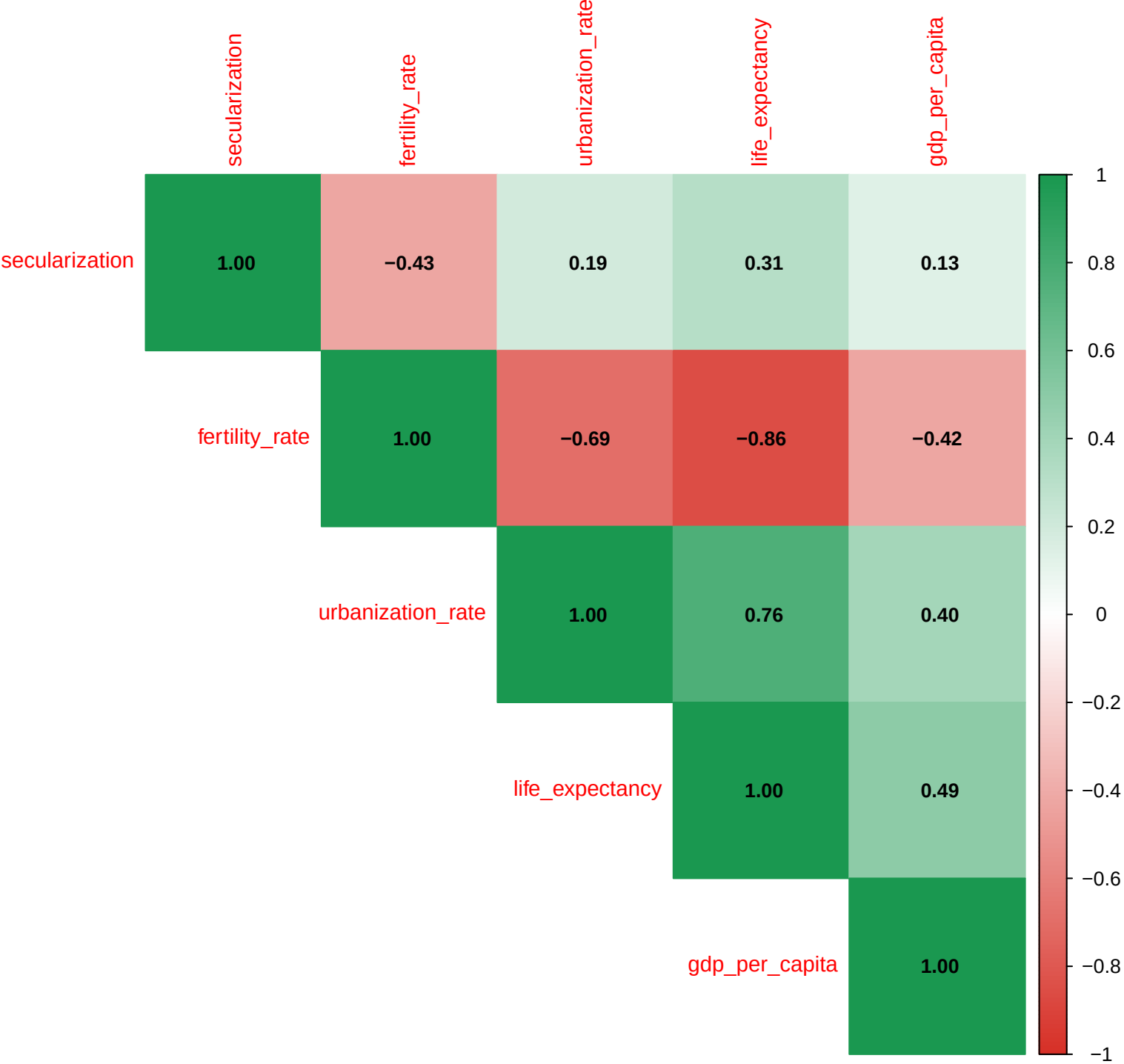
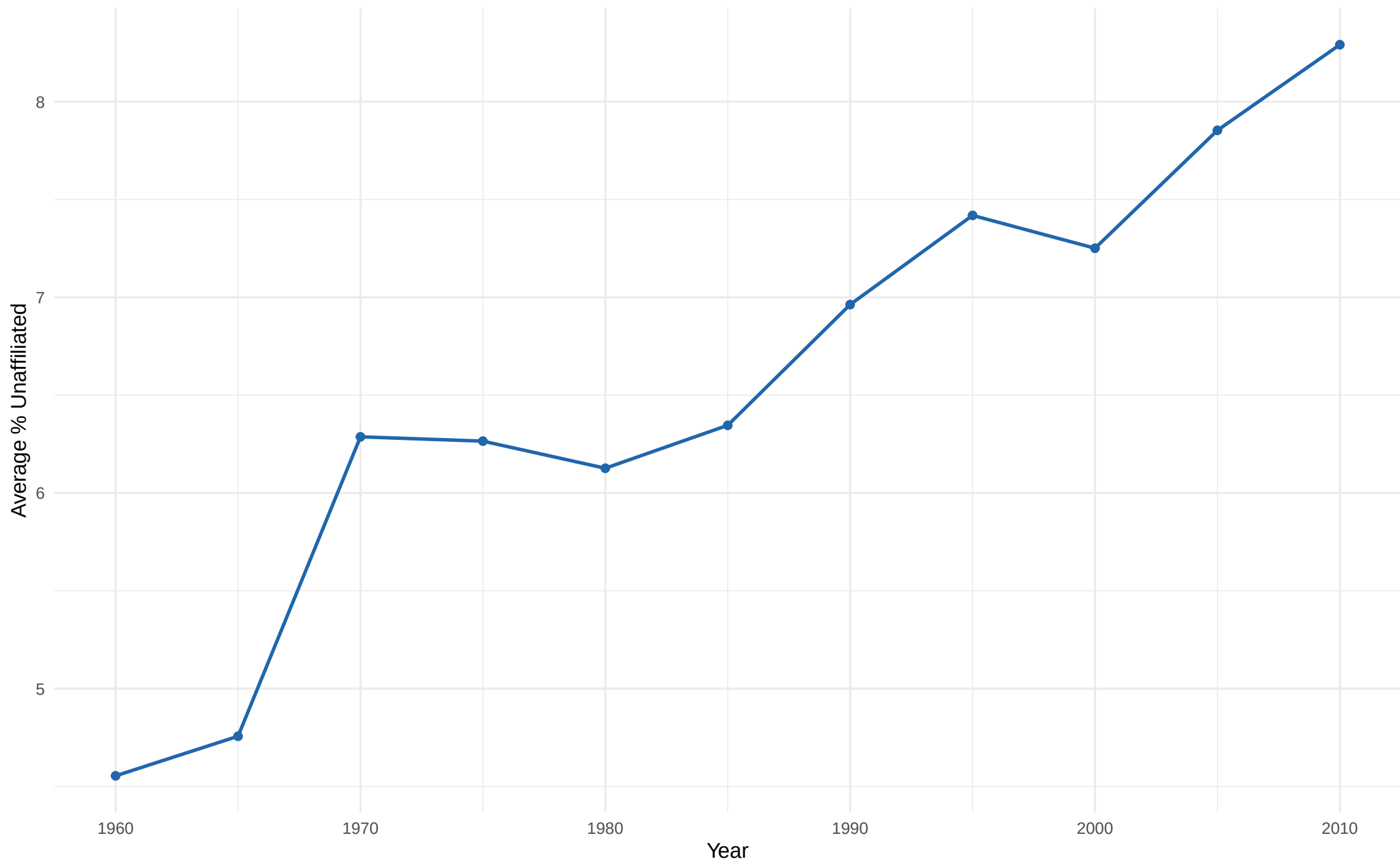


Correlation Matrix – Secularization & Development Indicators

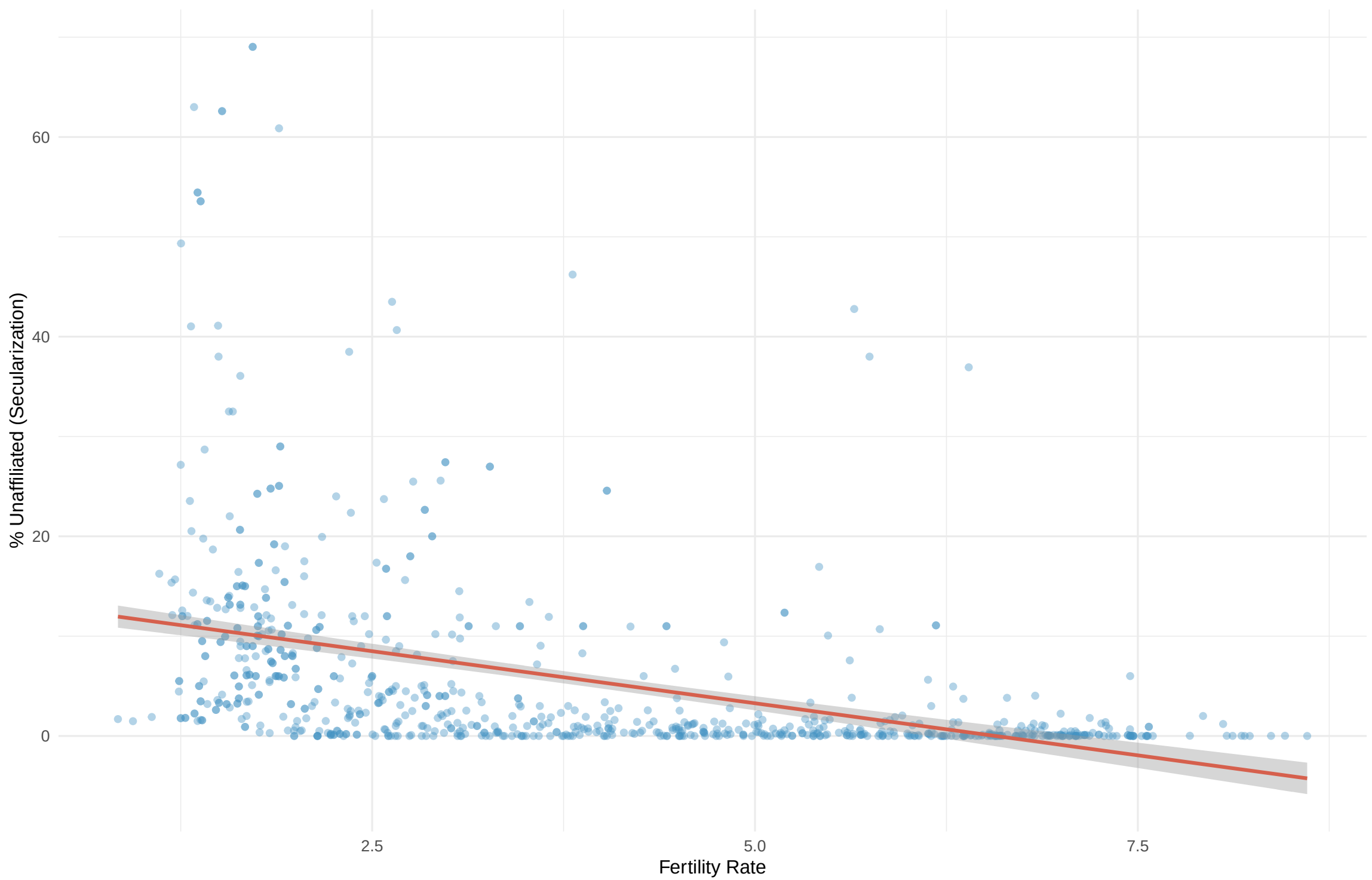


Global Secularization Trend (1945–2010)

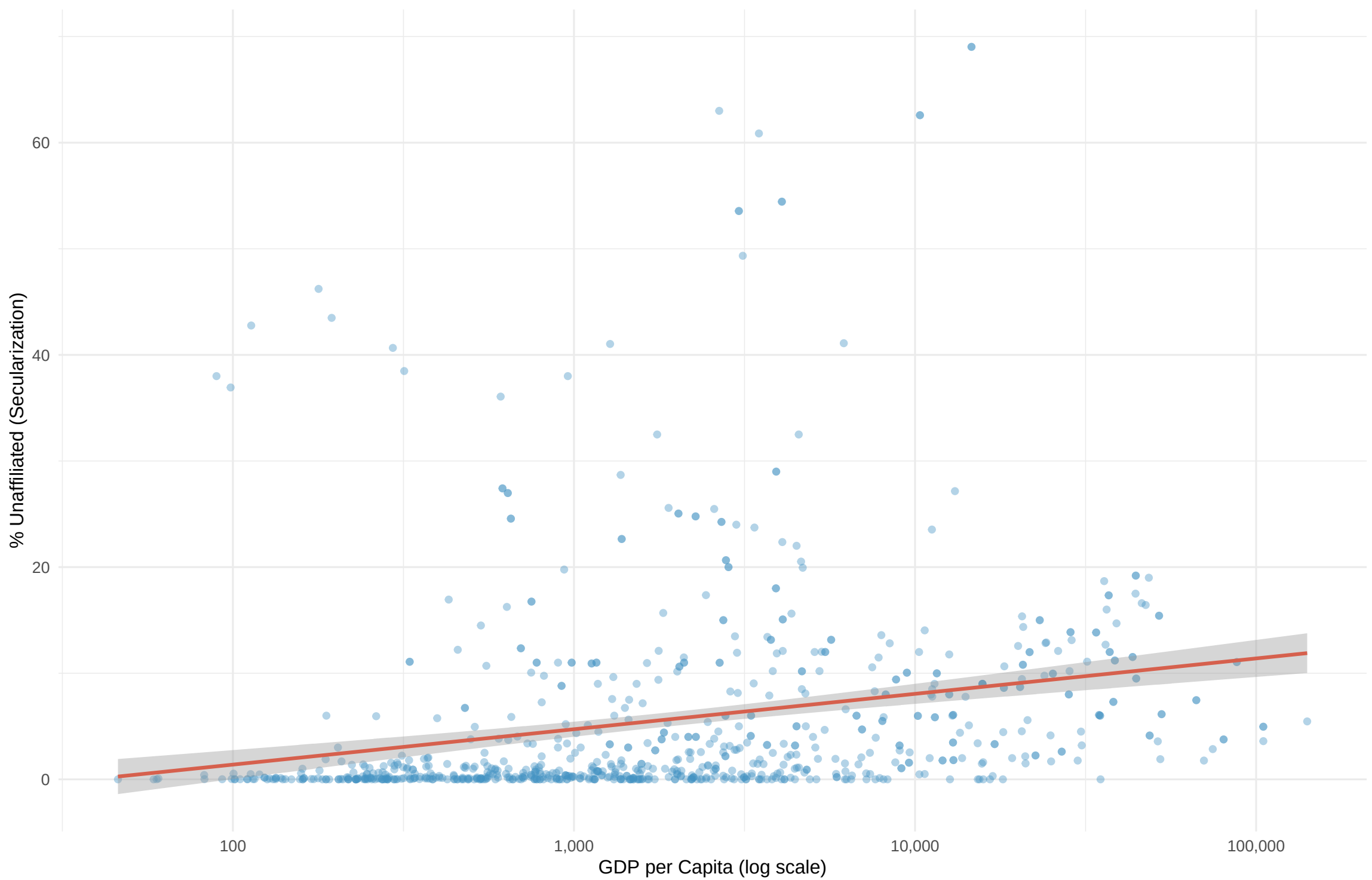
Average % population unaffiliated with any religion



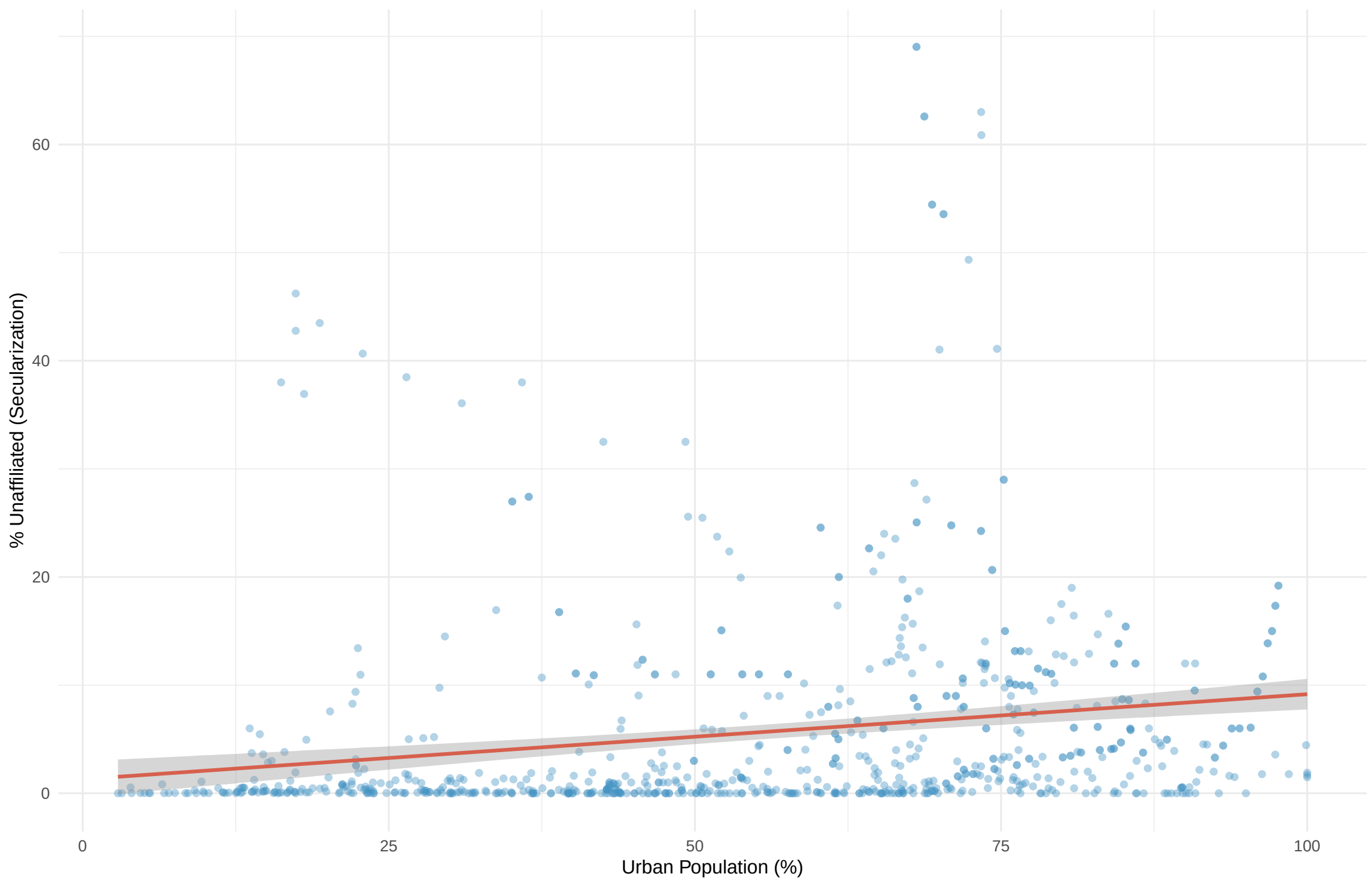
Fertility Rate vs Secularization



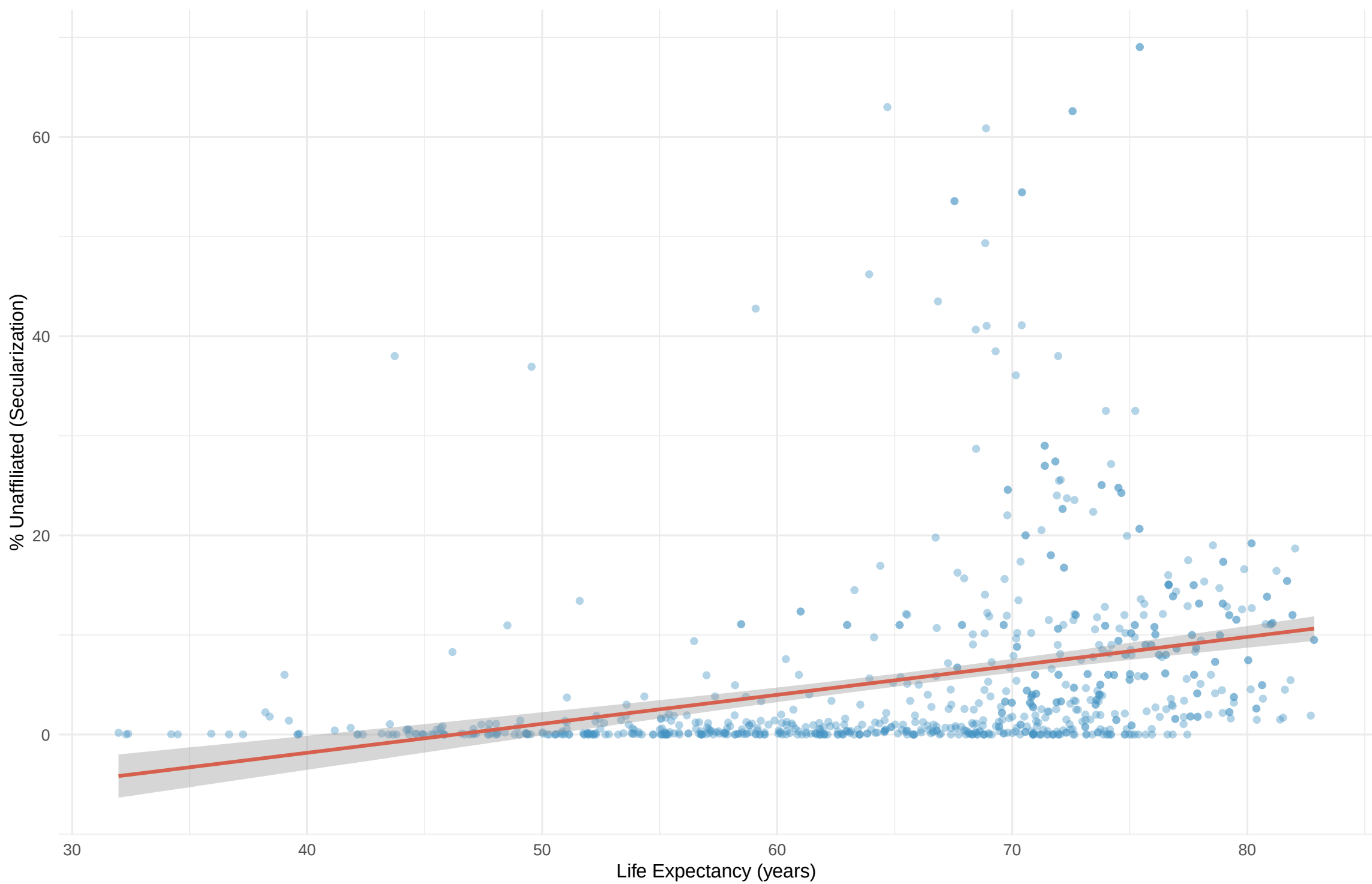
GDP per Capita (log scale) vs Secularization



Urban Population (%) vs Secularization

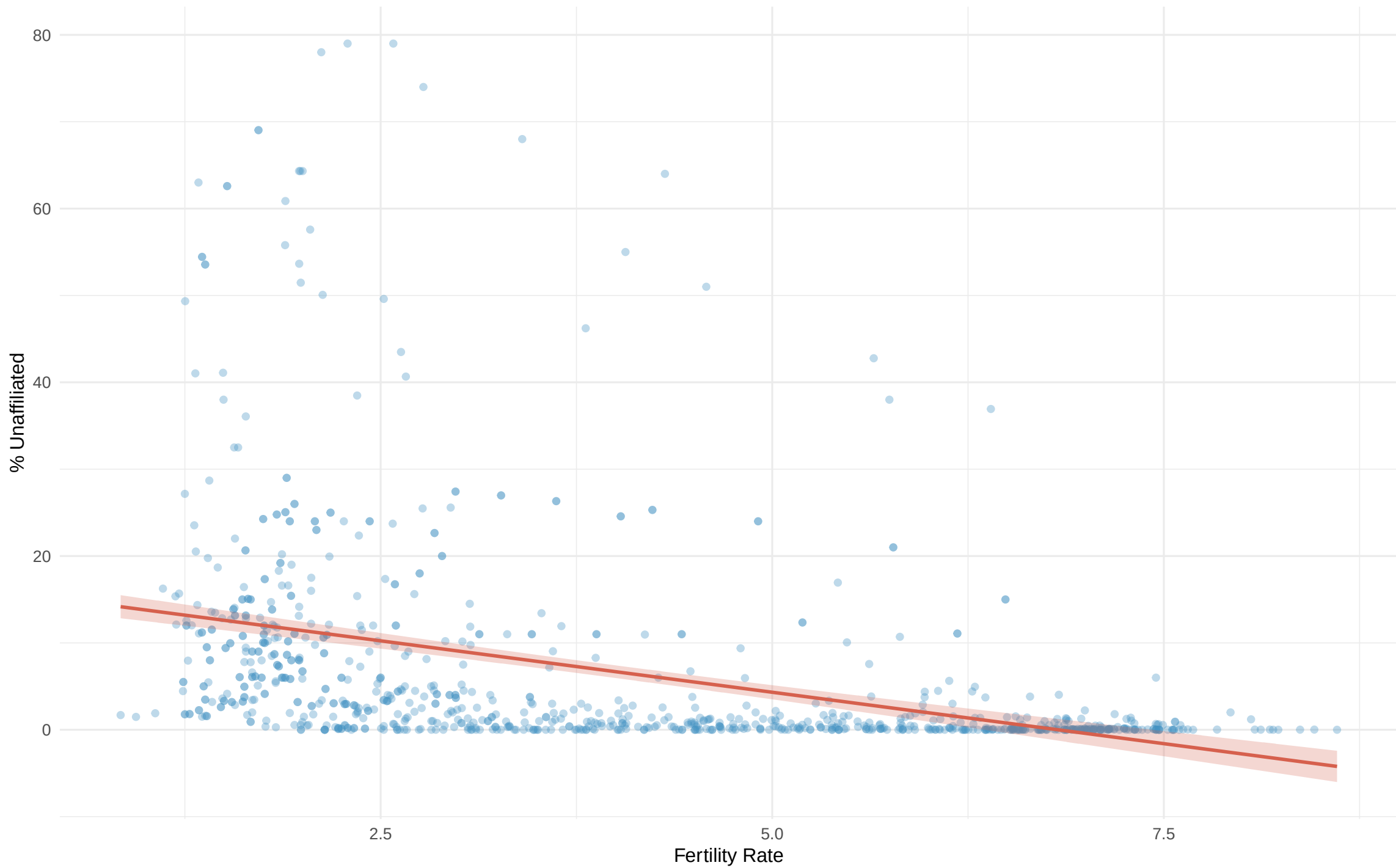


Life Expectancy (years) vs Secularization

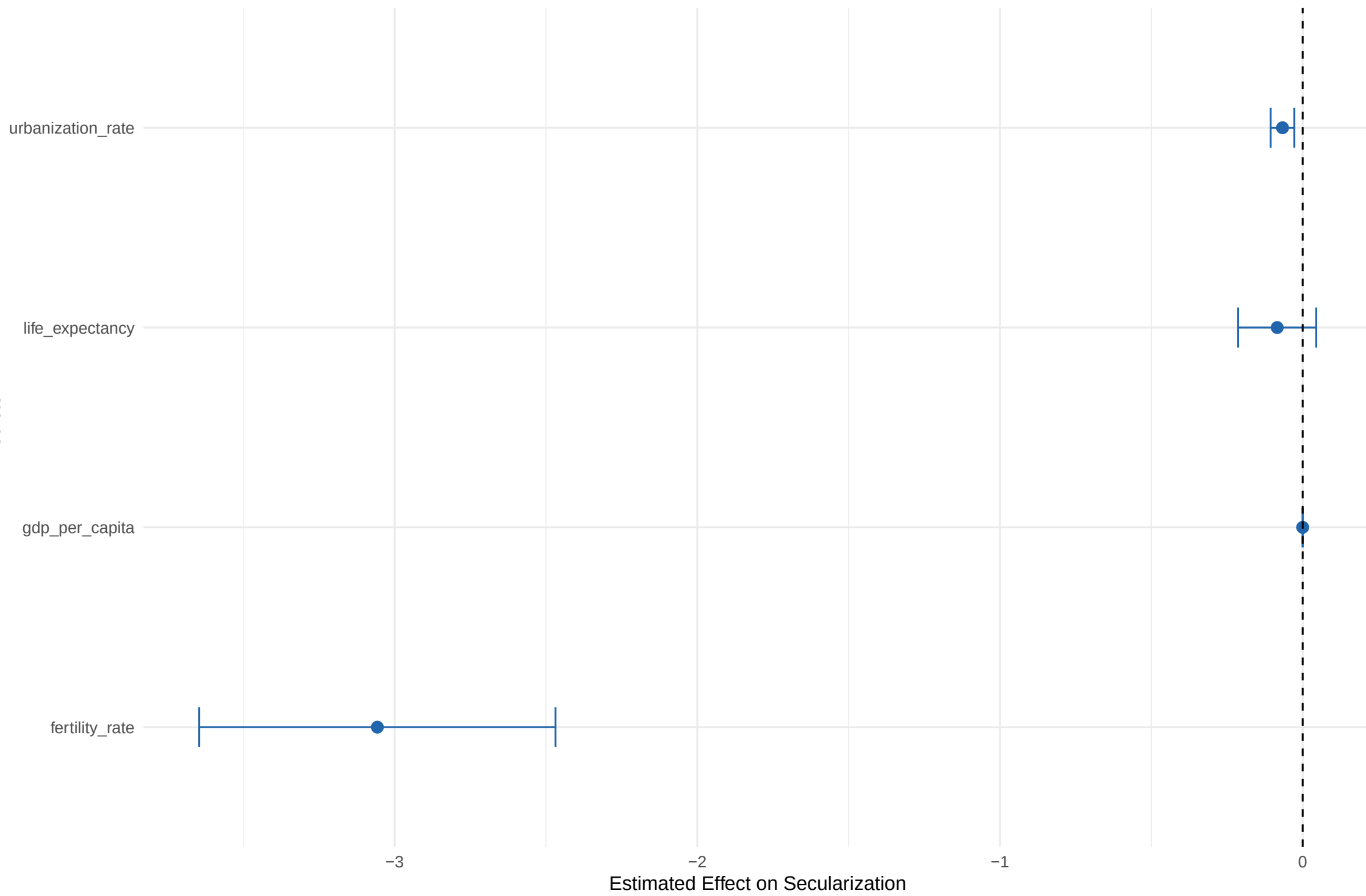


Fertility Rate vs Secularization with 95% CI

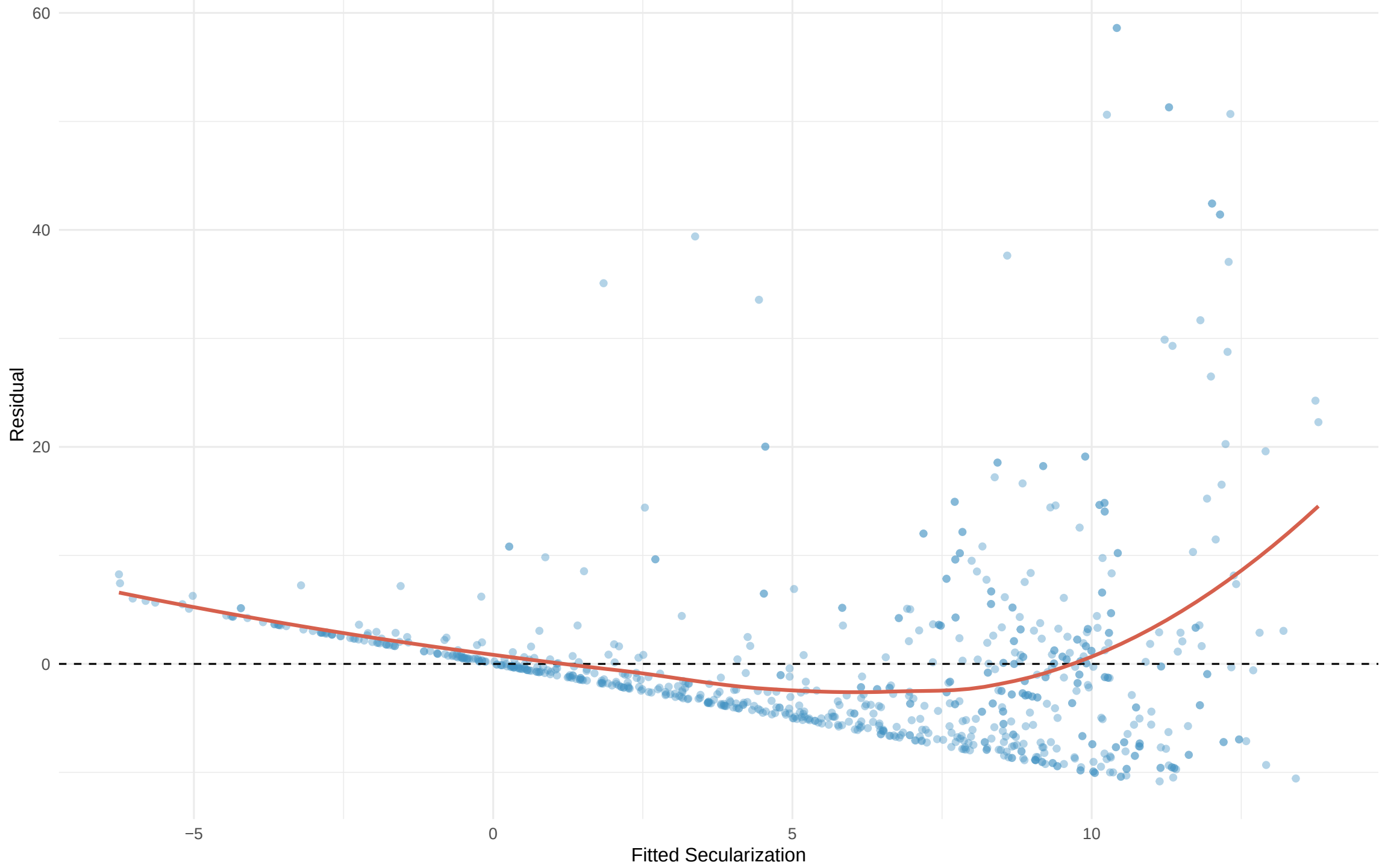
Simple OLS regression



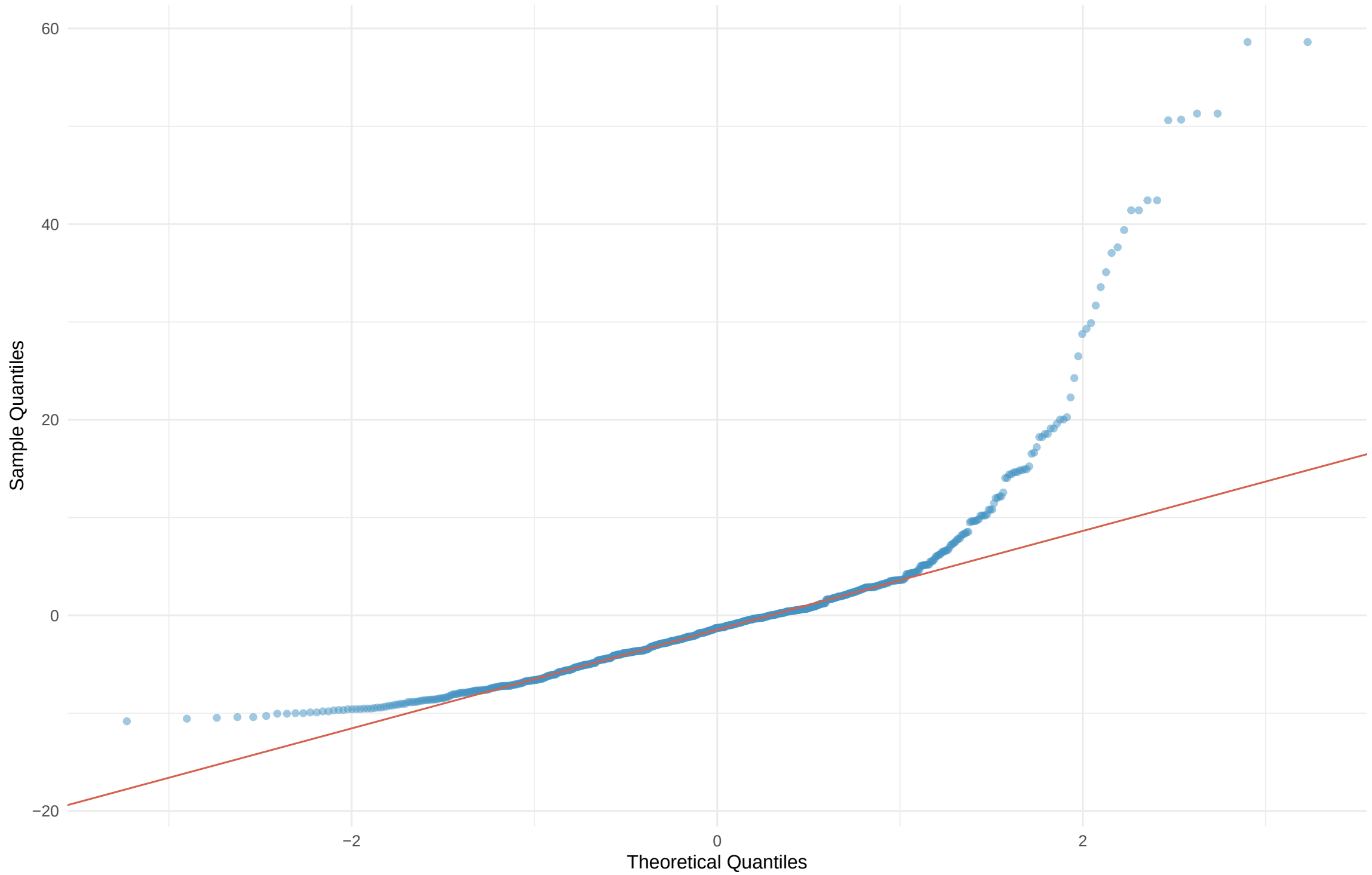
Multivariate OLS – Coefficient Estimates (95% CI)



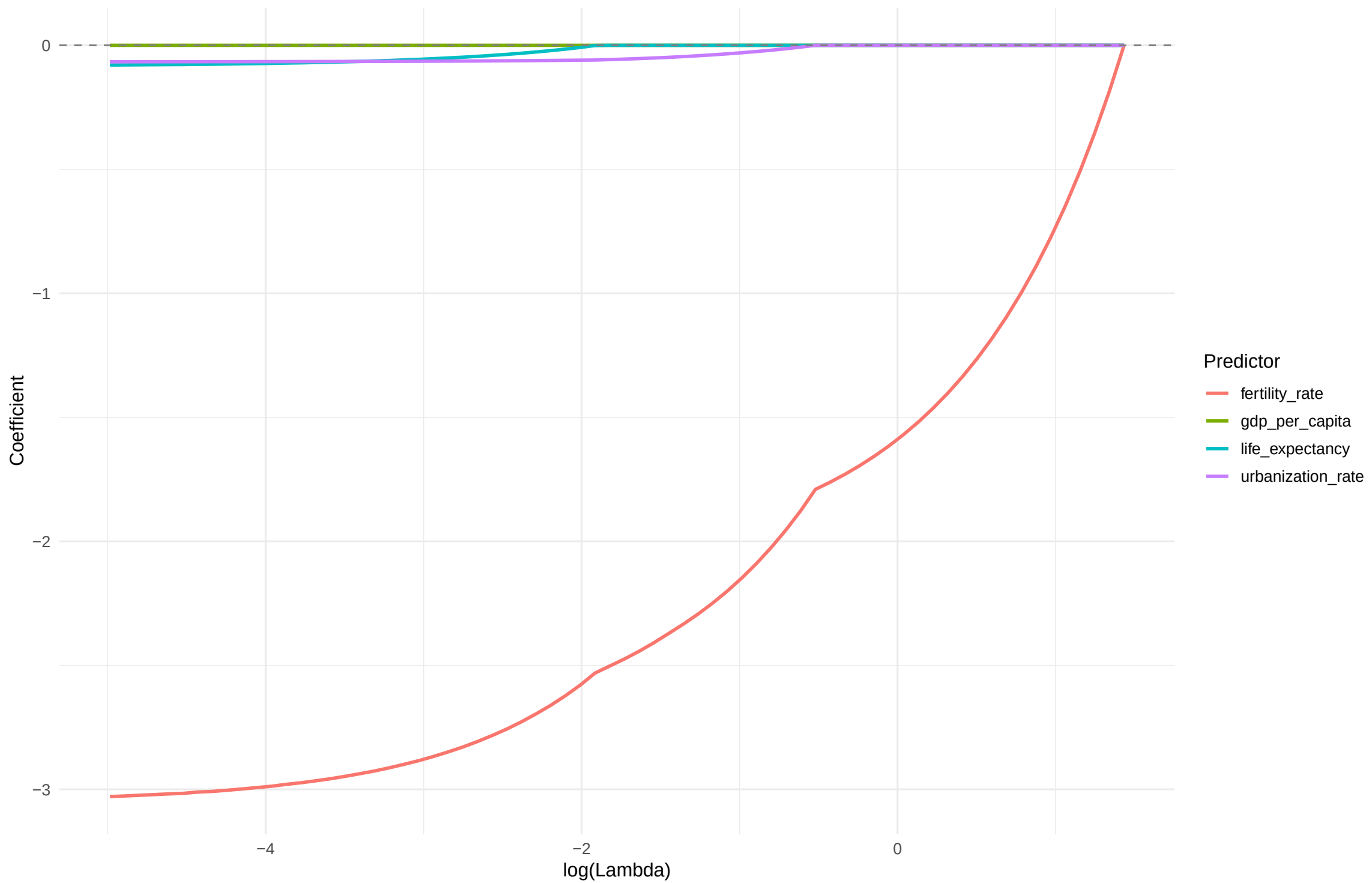
Residuals vs Fitted Values – Multivariate OLS



Normal Q-Q Plot of OLS Residuals

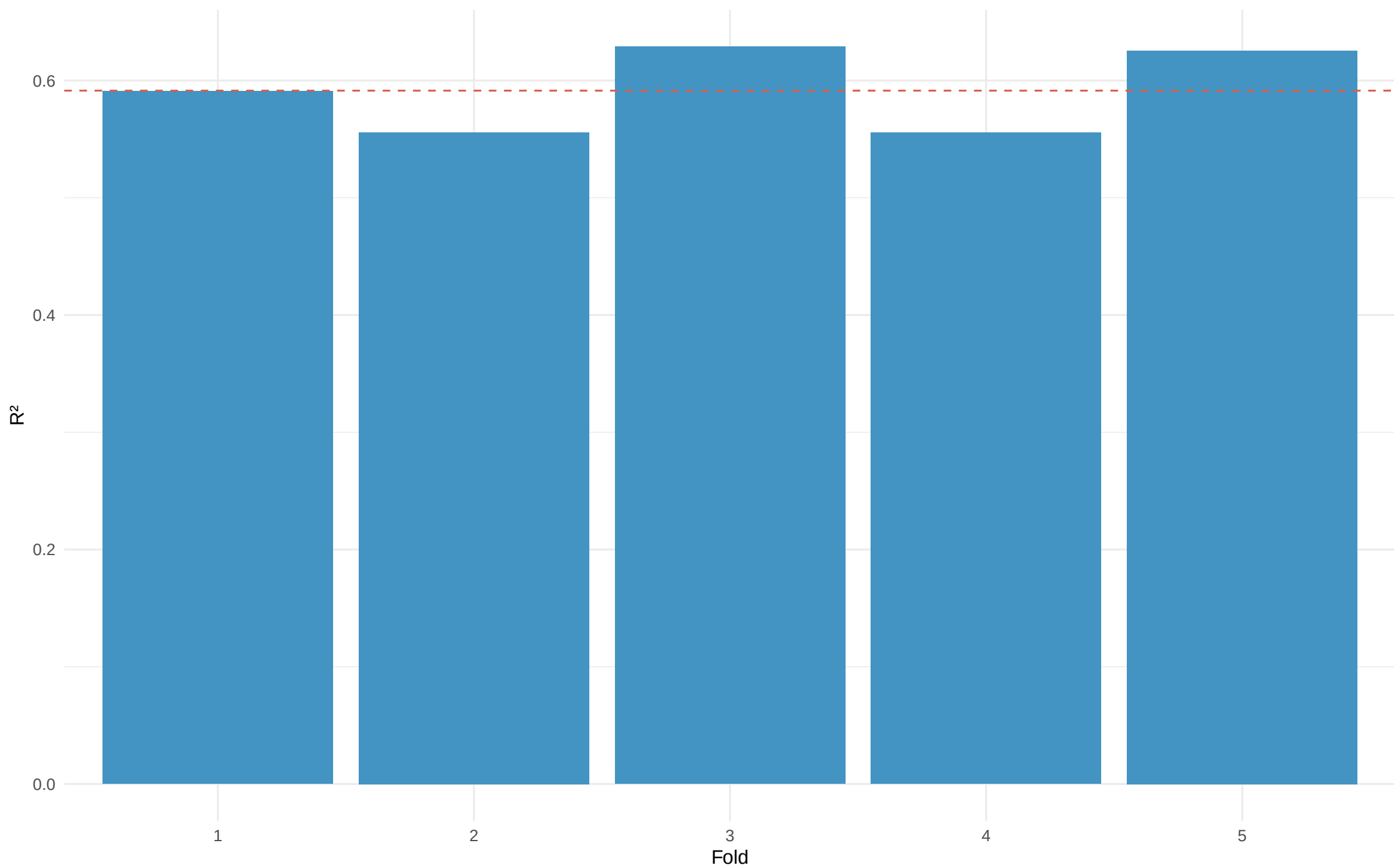


Lasso Coefficient Paths

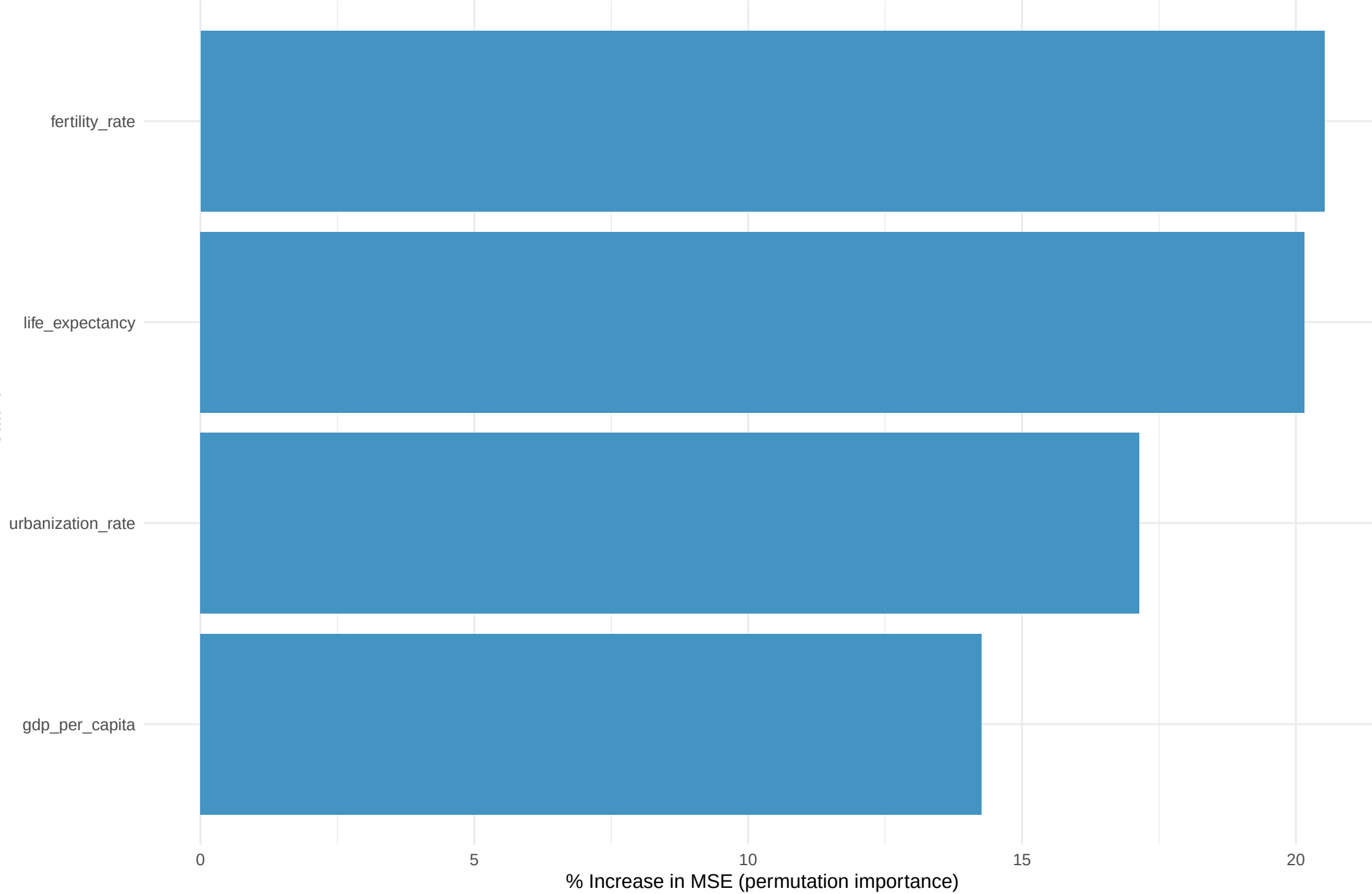


Random Forest R^2 Across 5 Cross-Validation Folds

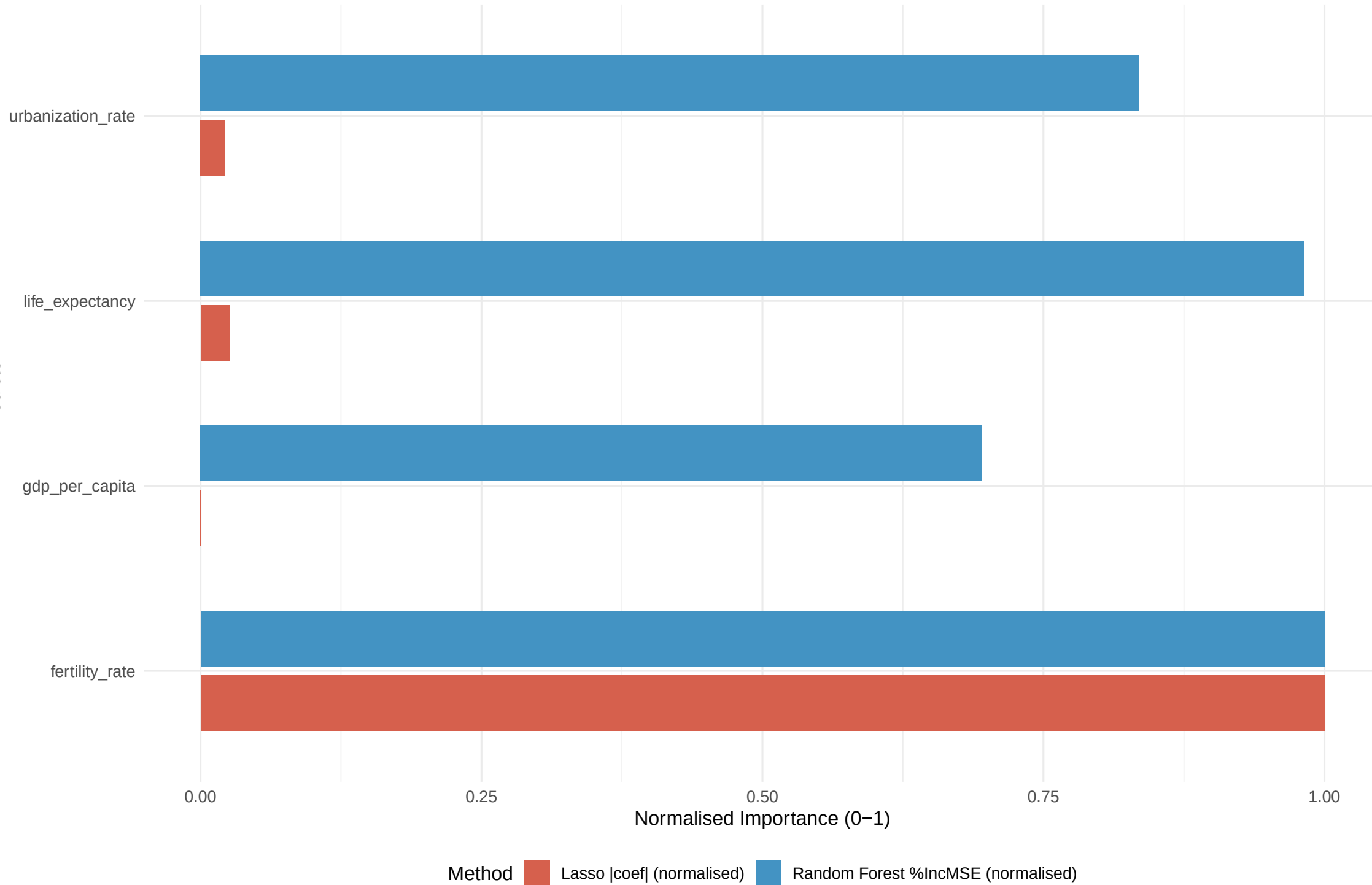
Mean $R^2 = 0.591$ (dashed line)



Random Forest – Feature Importance for Secularization

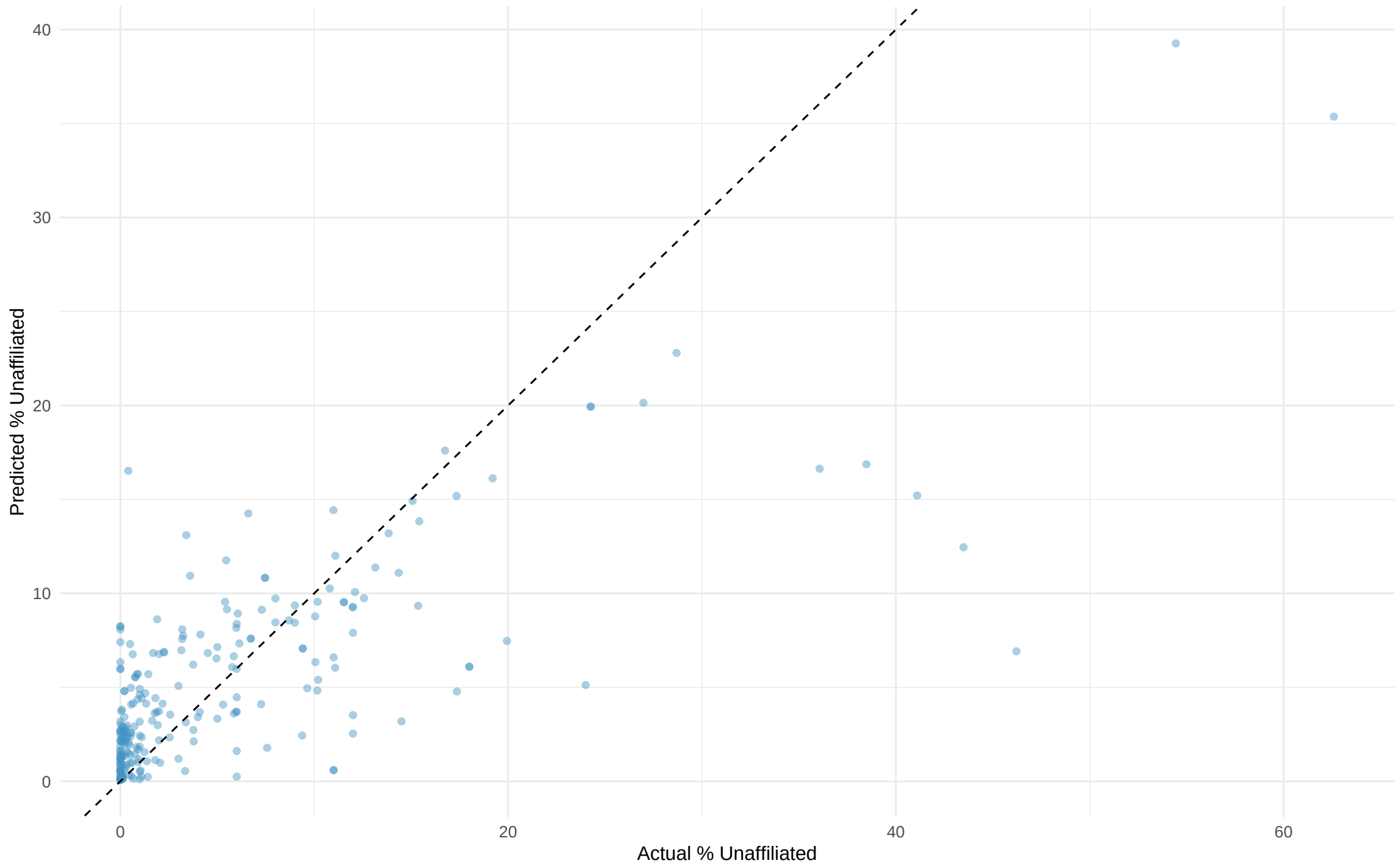


Predictor Importance: Lasso vs Random Forest



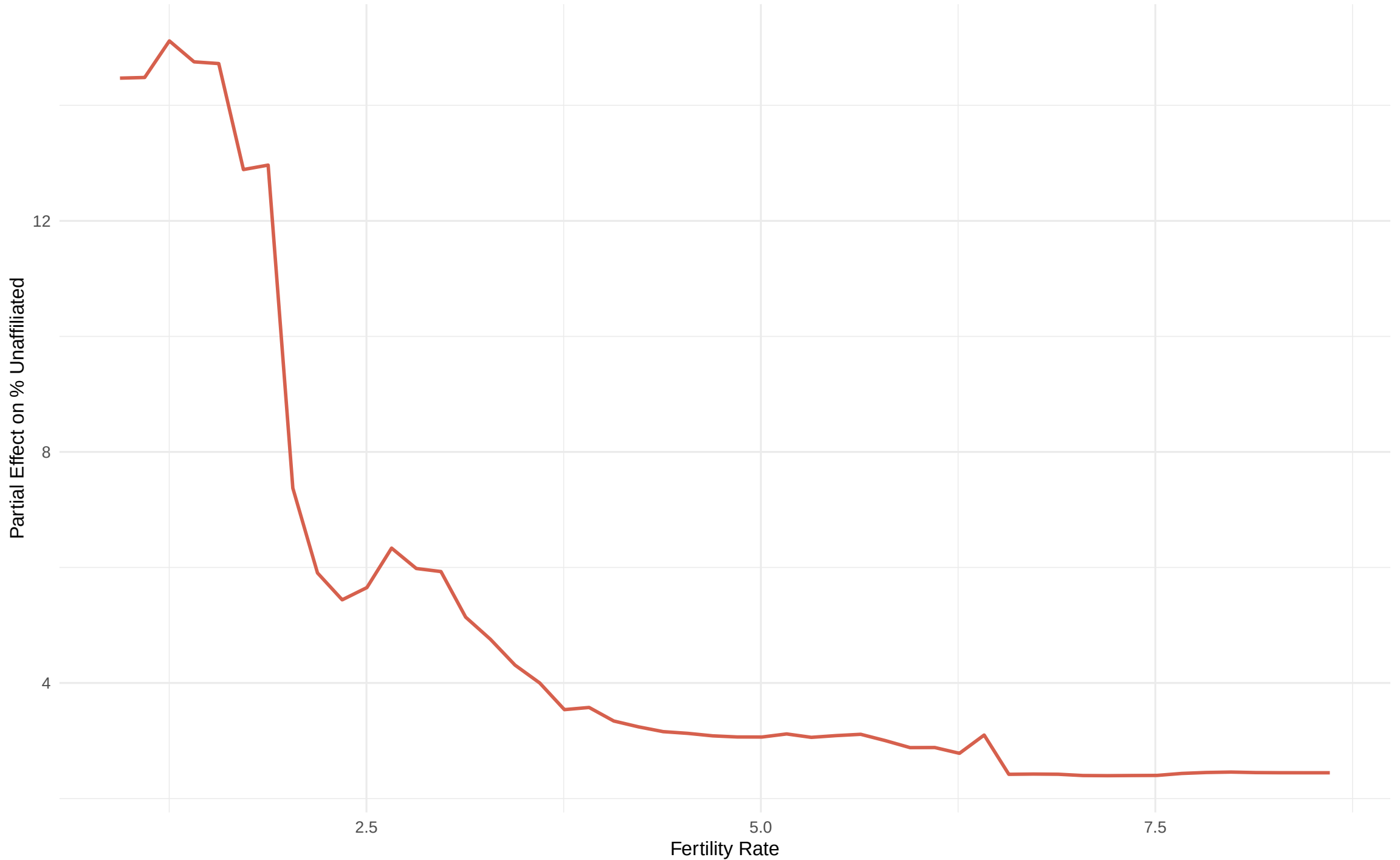
Random Forest: Actual vs Predicted Secularization

$R^2 = 0.561$ | RMSE = 6.04

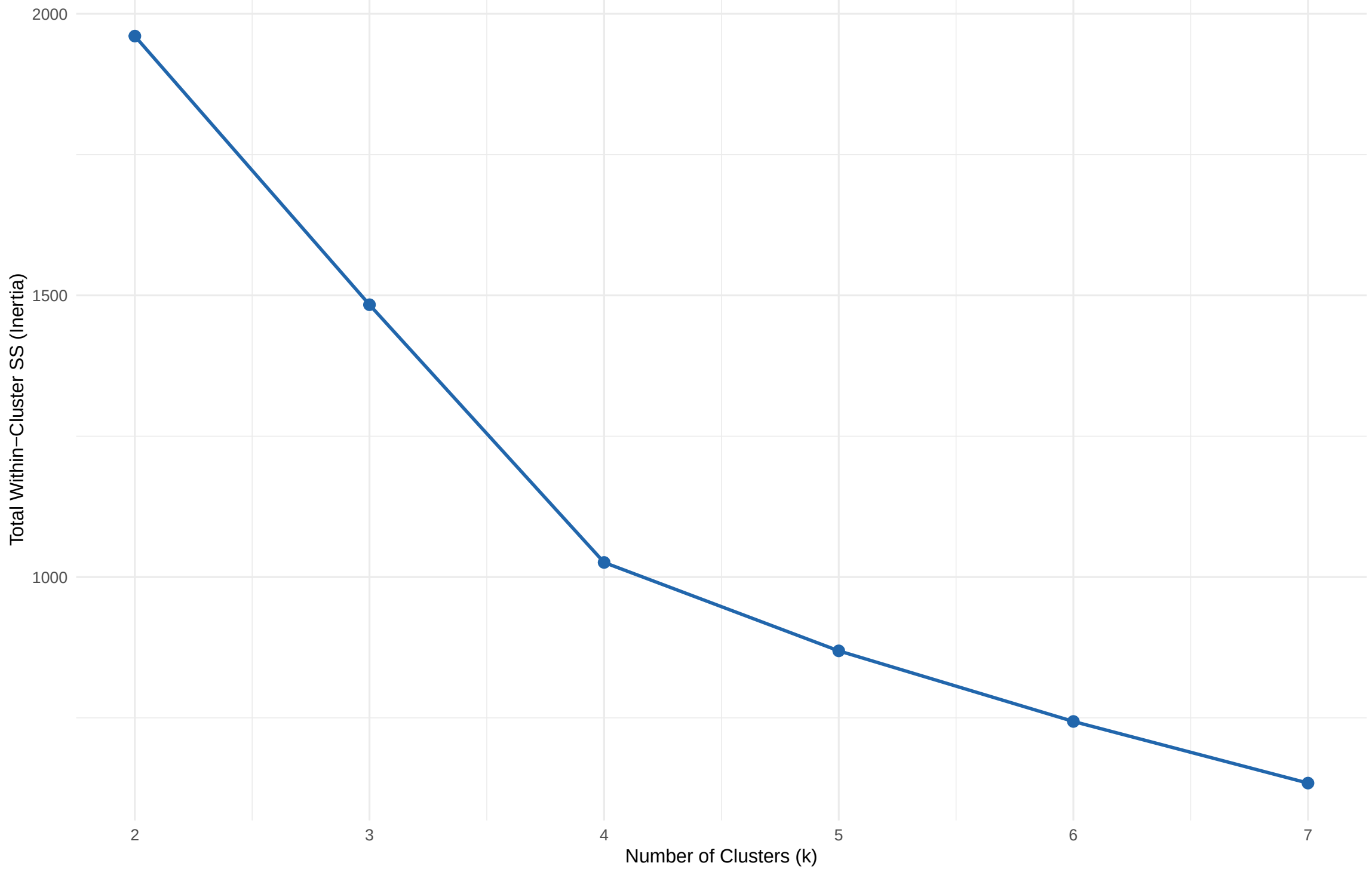


Partial Dependence Plot – Fertility Rate

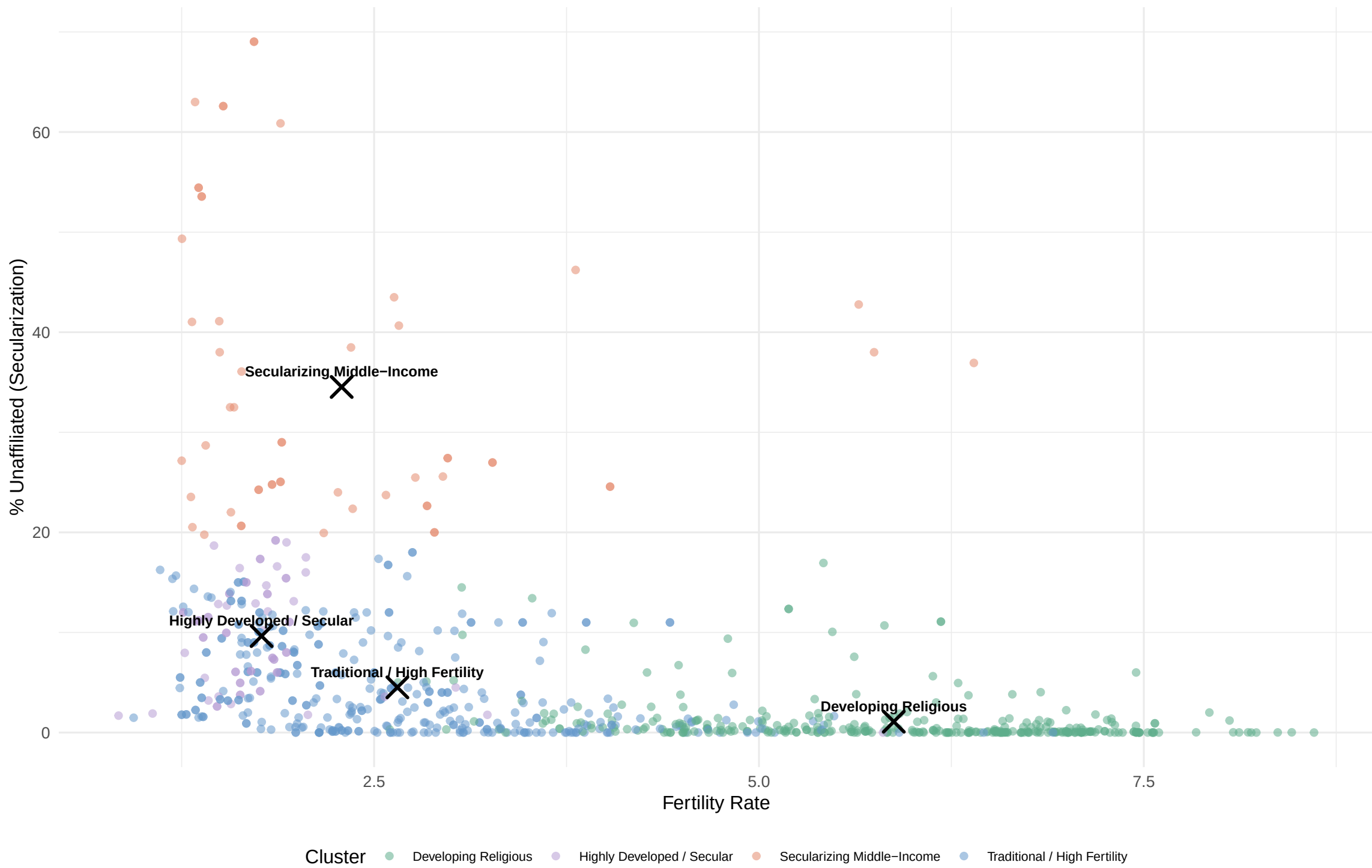
Effect of fertility rate on secularization (averaging over other predictors)



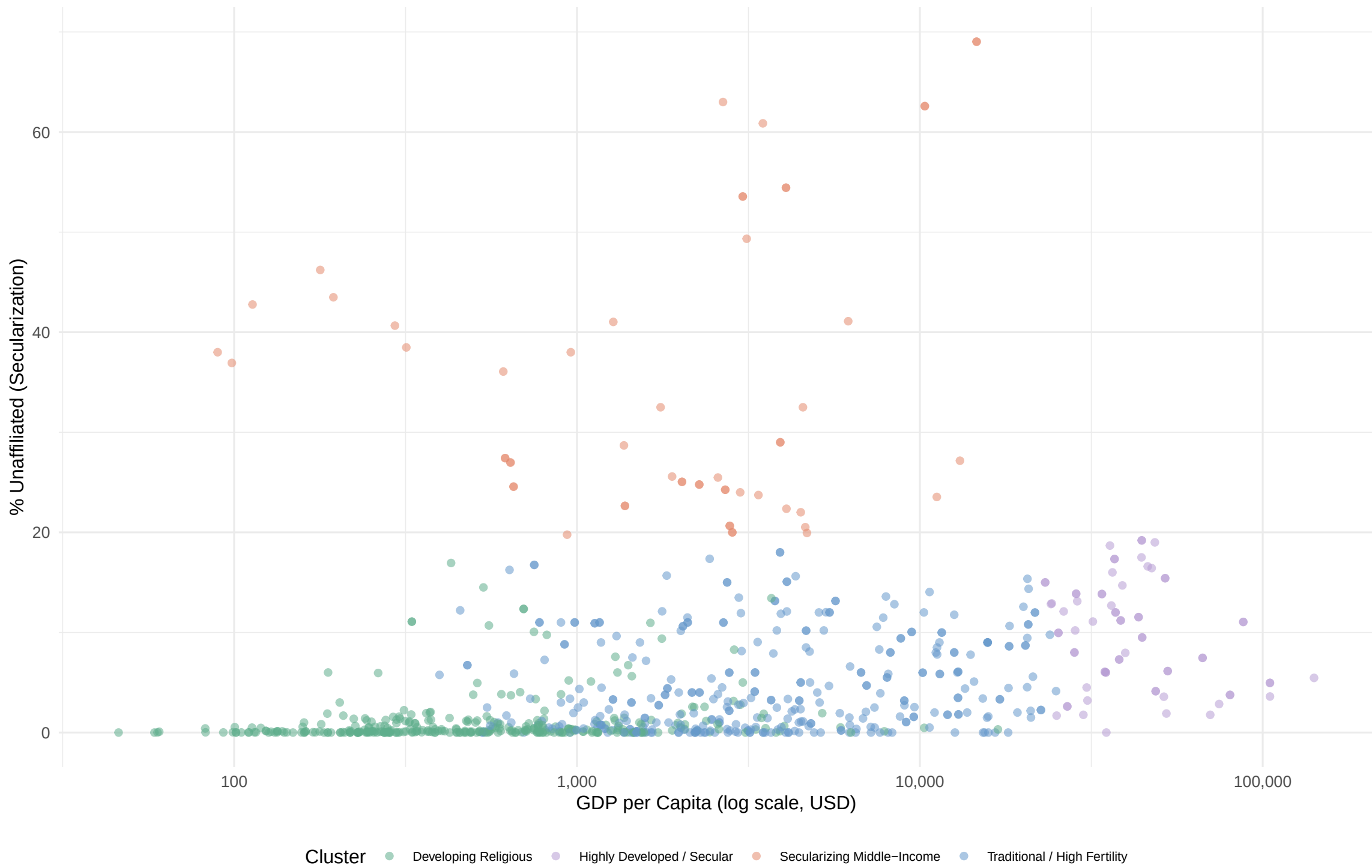
Elbow Method for Choosing Number of Clusters



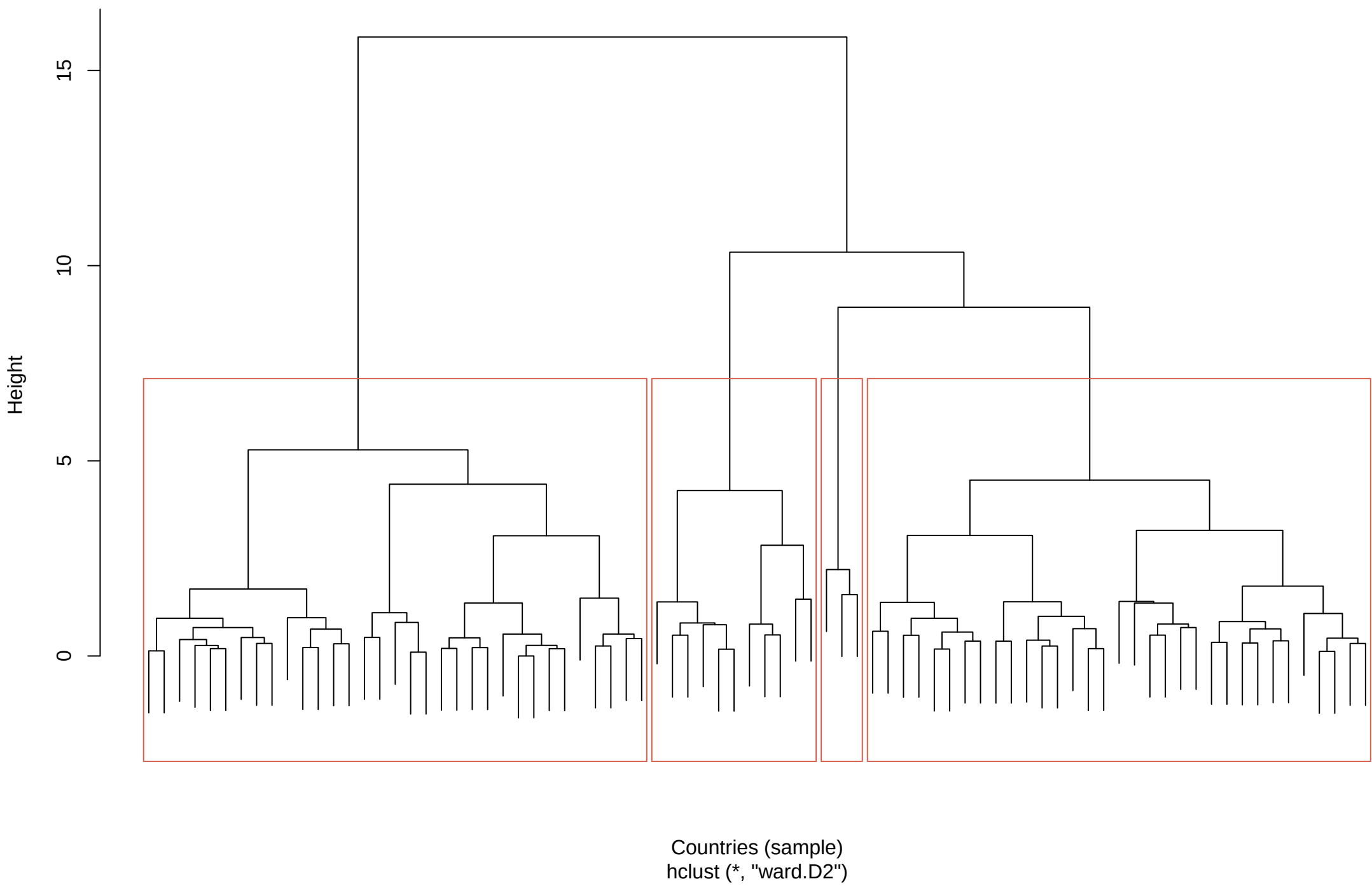
K-means Clusters: Fertility vs Secularization (k = 4)



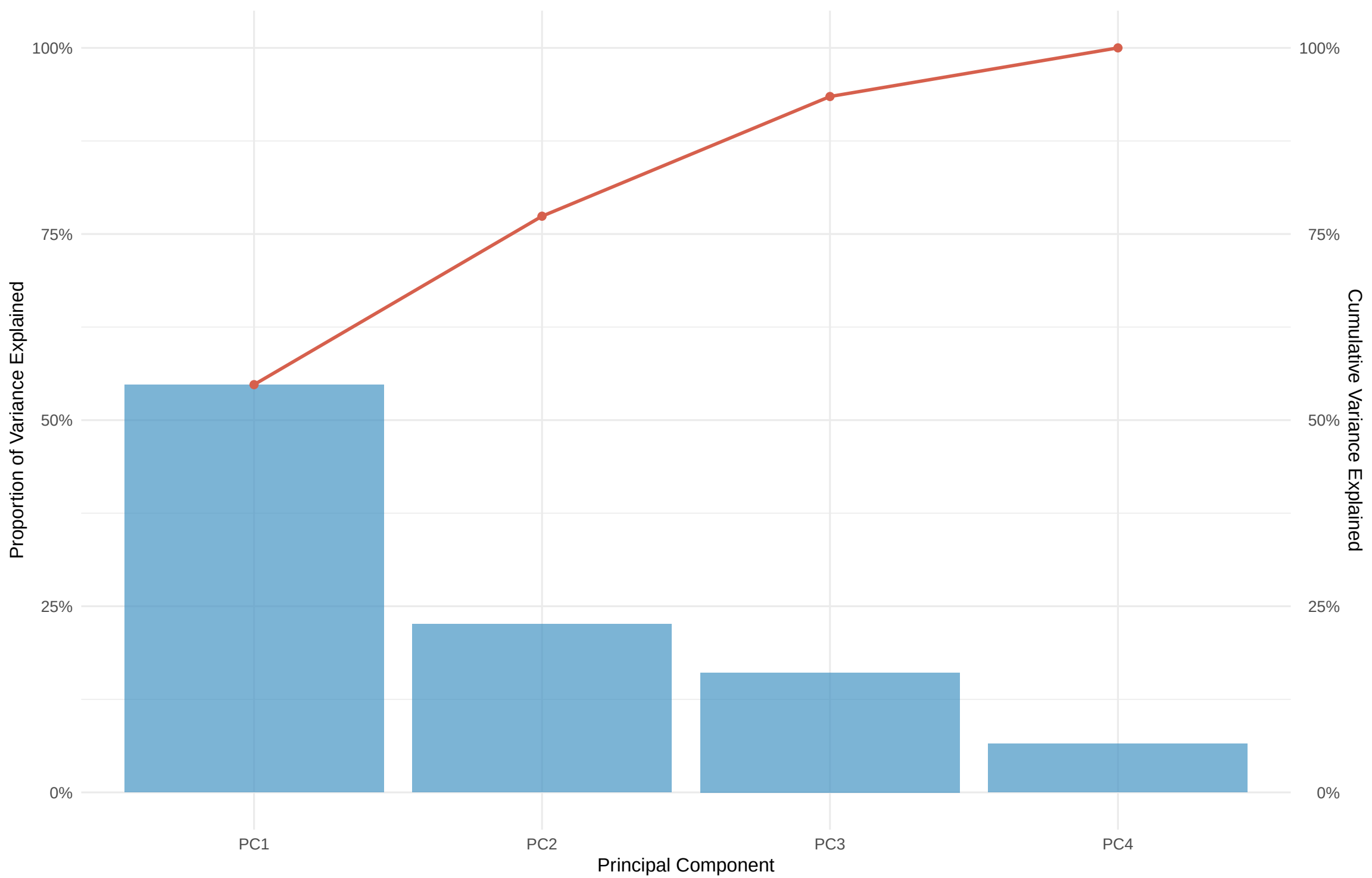
K-means Clusters: GDP per Capita vs Secularization (k = 4)



Hierarchical Clustering Dendrogram (Ward's method, n = 80 sample)



PCA Scree Plot



PCA Biplot – Countries Coloured by K-means Cluster

